D1.1 DNA replication

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-d1-1.html

Written for this site from the course textbook and class slides, with page references for each statement. The chapter is on pages 602–610 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Copying DNA

D1.1.1 Replication makes exact copies of DNA

p. 602

Replication produces copies with identical base sequences. It happens in the S phase, before a cell divides. It is needed for **reproduction**, so that offspring inherit base sequences, and for **growth** and **tissue replacement** in multicellular organisms, so that every new cell has a full set of genes.

Watch out: replication and cell division are separate events. DNA is copied first, and the cell divides later.

D1.1.2 Semi-conservative replication and base pairing

pp. 602–604

The two strands separate and each acts as a **template**. Each new molecule has **one old strand and one new strand**, so replication is **semi-conservative**.

Adenine pairs only with thymine, and cytosine only with guanine. Hydrogen bonds form only between complementary bases, so the template decides which nucleotide joins next. This is why copying is so accurate.

Base pairing makes copying accurate D1.1.2

Strand	Base sequence
Template	A T G A C C G C T
New strand	T A C T G G C G A

A pairs only with **T**; **C** pairs only with **G**. Only complementary bases form hydrogen bonds in the right position.
So the template decides which nucleotide joins next, and both copies match the original.

Key idea
Mistakes are rare, but when they happen they are the source of mutations (D1.3).

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D1.1.3 Helicase and DNA polymerase

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Enzyme	Role
Helicase	Unwinds the double helix and separates the strands, by breaking hydrogen bonds between bases
DNA polymerase	Links free nucleotides into a new strand, using the old strand as a template, by forming covalent bonds

Tools for working with DNA

D1.1.4 PCR and gel electrophoresis

pp. 605–608

The **polymerase chain reaction** copies a chosen section of DNA. Each cycle doubles the DNA, so 30 cycles give over a billion copies.

Step	Temperature	What happens
Denaturation	About 95°C	Strands separate
Annealing	About 40°C	Primers, which are short single strands of DNA, attach to mark the section
Elongation	About 72°C	Taq polymerase builds new strands from the primers

Taq polymerase comes from a bacterium living in hot springs, so it is not denatured at 95°C.

In **gel electrophoresis**, DNA is placed in wells and an electric current is applied. Negatively charged phosphate groups make DNA move towards the positive electrode. The gel acts as a sieve, so short fragments travel furthest. A DNA ladder of known lengths allows comparison.

Watch out: charge makes DNA move, but length is what separates the fragments.

D1.1.5 What these tools are used for

pp. 608–610

Some non-coding regions contain **short tandem repeats**. The number of repeats varies between people, so fragment lengths vary too. Many loci are compared, as a match at all of them is very unlikely by chance.

- **Forensics:** all bands from the crime scene sample must match the suspect. This is strong evidence that the DNA is theirs, but not in itself proof of guilt.
- **Paternity:** every band in a child comes from the mother or the father. Bands not found in the mother must be in the father.
- **Other uses:** diagnosing infections, and copying genes for research.

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